

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 9. Regular Expression

Lecture: 98. Special Character

Section 1: Lecture Summary

This lecture focuses on special characters in Python regular expressions (regex) that help define patterns for matching strings. Building upon quantifiers learned earlier, it introduces key special characters including square brackets `[]`, dot `.`, caret `^`, dollar sign `$`, and pipe `|`. Each character's meaning and usage is explained, such as defining sets or ranges of accepted characters, matching any single character, anchoring to beginning/end of strings, or expressing alternatives. The lecture then demonstrates how to combine these special characters with quantifiers to describe practical string patterns like dictionary words, capitalized names, variable names, times, and domain names. It emphasizes understanding the pattern components and their meanings before writing regex.

Section 2: Key Concepts and Explanations

- Square Brackets `[]` (Set of Possible Characters)

- Defines a set of allowed characters at a single position.
- Example: `[ABC]` matches any one character A, B, or C.
- Range notation: `[a-f]` matches any character from a to f.
- Multiple ranges can be combined: `[a-zA-Z0-9]` matches any alphanumeric character.
- Space inside brackets is considered a character; avoid spaces unless intended.
- Caret inside brackets `[^...]` negates the set, meaning match any character not in the brackets.

- Dot `.` (Any Character Except Newline)

- Matches any single character except newline.

- Example: ``.`` means one or more of any characters (excluding newline).
- To match a literal dot, escape it: `\.`

- **Caret `^` and Dollar `$` (Anchors)**

- `^` outside brackets asserts the match must start at the beginning of the string.
- `$` asserts the match must end at the end of the string.
- Inside brackets, `^` negates the character set (see above).

- **Pipe `|` (Alternation)**

- Allows matching one pattern or another.
- Example: `(com|org)` matches either "com" or "org".

- **Quantifiers**

- `+` means one or more repetitions.
- `*` means zero or more repetitions.
- `?` means zero or one repetition.

- **Combining Character Sets and Quantifiers**

- `[a-z]+` matches one or more lowercase letters.
- `[A-Z][a-z]*` matches a capital letter followed by zero or more lowercase letters.

- **Practical Pattern Examples**

- **Dictionary word:** `[a-z]+` any lowercase word.
- **Capitalized word:** `[A-Z][a-z]*` first letter capital, rest lowercase.
- **First and Last name:** `[A-Z][a-z]+ [A-Z][a-z]*` two words separated by space, each starting with capital.

- **Variable name:** ``[a-zA-Z_][a-zA-Z0-9_]`` start with letter or underscore, followed by letters, digits, or underscores.
 - **Time (24-hour format):** ``[0-9]:[0-5][0-9]`` hours from 00–29 (basic check), minutes 00–59.
 - **Domain name:** ``[a-zA-Z0-9]+\.(com|org|net)$`` alphanumeric domain ending with common extensions.
-
- **Escape sequences**
 - Backslash ``\`` used to escape special characters.
 - For example, ``\.`` to match a literal dot.

Section 3: Example Code and Use Cases

Example 1: Any lowercase word (one or more letters)

Example 2: Capitalized words (first capital, others lowercase)

Example 3: First name and last name pattern

Example 4: Variable name pattern

Example 5: Time in 24-hour format (basic check)

Example 6: Domain name pattern (common extensions)

Example 7: Matching a string starting with "http"

```
import re

pattern = r'[a-z]+'
print(re.match(pattern, 'board')) # Matches 'board'

pattern = r'[A-Z][a-z]*'
print(re.match(pattern, 'Abdul')) # Matches 'Abdul'

pattern = r'[A-Z][a-z]+ [A-Z][a-z]*'
print(re.match(pattern, 'Abdul Bari')) # Matches full name

pattern = r'[a-zA-Z_][a-zA-Z0-9_]*'
print(re.match(pattern, 'item1')) # Matches variable name
```

```
pattern = r'[012][0-9]:[0-5][0-9]'\nprint(re.match(pattern, '11:30')) # Matches time\n\npattern = r'[a-zA-Z0-9]+\\.(com|org|net)$'\nprint(re.match(pattern, 'udemy.com')) # Matches domain\n\npattern = r'^http.+'\nprint(re.match(pattern, 'http://google.com')) # Matches URL start
```

Each pattern corresponds to a discussed real-world validation scenario, illustrating how special characters and quantifiers are combined to define string formats.

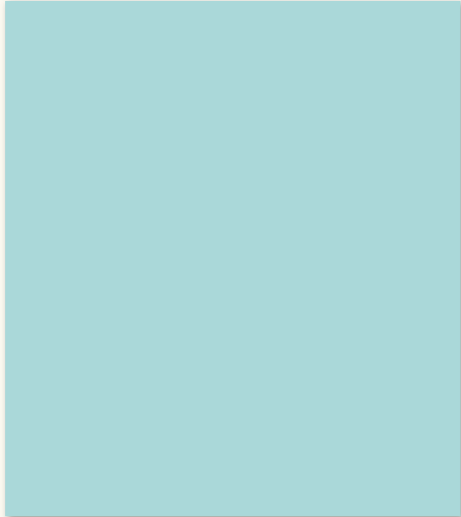
Section 4: Key Takeaways

- Square brackets `[ ]` are used to specify sets or ranges of allowed characters at one position.
- `^` and `\$` anchor the regex to the beginning and end of the string.
- Dot `.` matches any single character except newline, and must be escaped to match a literal dot.
- Pipe `|` provides a choice between patterns.
- Quantifiers modify how many times the preceding element can repeat: `+`, `\*`, and `?`.
- Regex patterns can be combined to define practical validations, e.g., names, variable names, time formats, and domain names.
- Understanding basic building blocks helps create powerful regexes to match complex patterns.
- Escape special characters if they should match literally.
- Practice writing and testing regex patterns with real examples to gain proficiency.

Lecture in Slides

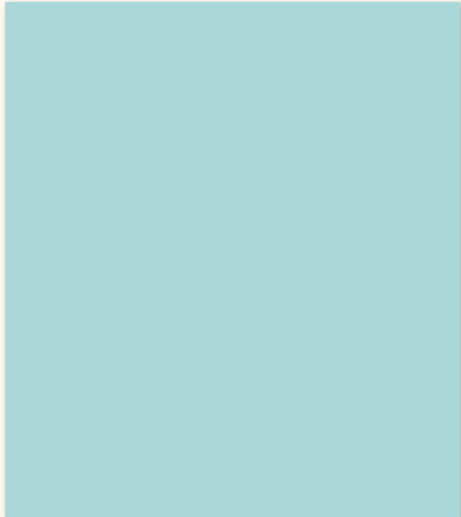
Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





Regular Expressions

Special Characters



Special Characters

Special Characters

Special Characters

[...] set of possible characters

Special Characters

[...]

set of possible characters

[^...]

all characters except in bracket

Special Characters

[...]

set of possible characters

[^...]

all characters except in bracket

.

any character except newline

Special Characters

[...]

set of possible characters

[^...]

all characters except in bracket

.

any character except newline

^

beginning of a String

Special Characters

[...]

set of possible characters

[^...]

all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

Special Characters

[...]

set of possible characters

[^...]

all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[]'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[abc]'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[abc]'

'a' **'b'** **'c'**

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[abcdef]'

'a' **'b'** **'c'**

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[abcdef]'

'a' **'b'** **'c'**

'd' **'e'** **'f'**

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-f]'

'a' **'b'** **'c'**

'd' **'e'** **'f'**

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]'

'a'

'b'

'c'

'd'

'e'

'f'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]'

'a'

'b'

'c'

'd'

'e'

'f'

'g'

'h'

'i'

.

.

.

.

.

.

'x'

'y'

'z'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]'

any single **LOWER** case alphabet

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]'

any single **LOWER** case alphabet

Lower case

Upper case

Digits

Special Chars

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-zA-Z]'

any single **LOWER** or **UPPER** case alphabet

✓
Lower case

✓
Upper case

Digits

Special Chars

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-zA-Z0-9]'

any single LOWER case or UPPER case or DIGIT

✓ ✓ ✓
Lower case Upper case Digits Special Chars

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]+'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]+'

'b'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]+'

'bo'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]+'

'boar'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]+'

'board'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[a-z]+'

'board' **'pen'**

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]+'

'board'

'pen'

'keys'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[a-z]+'

any **DICTIONARY** word

'board'

'pen'

'keys'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z]'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z][a-z]'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z][a-z]+'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z][a-z]+'

first letter is **UPPER** case

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z][a-z]+'

first letter is **UPPER** case

any 2 letter **DICTIONARY** word

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'[A-Z][a-z]*'

first letter is **UPPER** case

any **DICTIONARY** word

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

'[a-z]+'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

'[^a-z]+'

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[^a-z]+'

Strings with UPPER case or DIGIT or SPECIAL char

Special Characters

[...]	set of possible characters
[^...]	all characters except in bracket
.	any character except newline
^	beginning of a String
\$	end of a String
R S	R or S

pattern

'[^a-z]+'

Strings with UPPER case or DIGIT or SPECIAL char

Lower case

Upper case

Digits

Special Chars

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

. any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

['']

Special Characters

[...]

 set of possible characters

[^...]

 all characters except in bracket

.

 any character except newline

^

 beginning of a String

\$

 end of a String

R|S

 R or S

pattern

‘.+’

Special Characters

[...]

 set of possible characters

[^...]

 all characters except in bracket

.

 any character except newline

^

 beginning of a String

\$

 end of a String

R|S

 R or S

pattern

‘.+’

Strings with

LOWER case

UPPER case

DIGIT

SPECIAL char

Except newline

✓

Lower case

✓

Upper case

✓

Digits

✓

Special Chars

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

pattern

'http'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^ beginning of a String

\$ end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

^http

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

^http

http

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

‘^http.+’

‘http’

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

‘^http.+’

‘http://google.com’

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

‘^http.+’

‘http://google.com’

‘use http protocol’

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

‘^http.+’

‘http://google.com’

~~**‘use http protocol’**~~

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

'com\$'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

'com\$'

'udemy.com'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S

R or S

pattern

'.+com\$'

'udemy.com'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S R or S

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S R or S

pattern

'com|org'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S R or S

pattern

'com|org'

'udemy.com'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S R or S

pattern

'com|org'

'udemy.com'

'study.org'

Special Characters

[...] set of possible characters

[^...] all characters except in bracket

.

any character except newline

^

beginning of a String

\$

end of a String

R|S R or S

?+*{m}[...][^...].^\$R/S

?+*{m}[...][^...].^\$R/S

First Name and Last Name

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z]'

?

+

*

{m}

[...]

[^...]

.

^

\$

R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]'

?

+

*

{m}

[...]

[^...]

.

^

\$

R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

?

+

*

{m}

[...]

[^...]

.

^

\$

R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+' [A-Z][a-z]+'

Variable Name

?

+

*

{m}

[...]

[^...]

.

^

\$

R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

?

+

*

{m}

[...]

[^...]

.

^

\$

R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'['']'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-z]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z_0-9]*'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z_0-9]*'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5] '

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

?+*{m}[...][^...].^\$R|S

First Name and Last Name

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'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

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'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

?+*{m}[...][^...].^\$R|S

First Name and Last Name

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'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

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Time

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'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

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Time

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'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

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'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z0-9]'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

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'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z0-9]+\.' \$'

?+*{m}[...][^...].^\$.R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z0-9]+\.(com)\$'

?+*{m}[...][^...].^\$.R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z0-9]+\.(com|org)\$'

?+*{m}[...][^...].^\$R|S

First Name and Last Name

Abdul Bari

'[A-Z][a-z]+ [A-Z][a-z]+'

Variable Name

item1

'[a-zA-Z_][a-zA-Z0-9_]*'

Time

11:30

'[012][0-9]:[0-5][0-9]'

Domain Name

udemy.com

'[a-zA-Z0-9]+\.(com|org|net)\$'